

ANALYSIS REPORT

COMPLIANCE REPORT

Elliptic Bitcoin Transaction Network | Fraud Pattern Investigation

Submitted To: **Head of Compliance**

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Dataset: <https://www.kaggle.com/datasets/ellipticco/elliptic-data-set>

Reference Bodies: **Central Bank of Nigeria (CBN) / NFIU**

Tools: **Google Colab, Pandas, NumPy, NetworkX, Matplotlib, Plotly**

Date: **May 2026**

1. Executive Summary

This report presents the outcome of a graph-based compliance investigation conducted on the Elliptic Bitcoin Transaction Dataset containing over 203,000 Bitcoin transactions across 49 time steps. The analysis identified structurally suspicious transaction hubs, layering behavior, and fan-out transaction patterns associated with possible money laundering activity.

A focused forensic subgraph of 5,459 nodes and 6,478 edges was extracted for investigation. Community analysis identified 74 transaction clusters, including several communities dominated by unknown or unverified entities. Centrality analysis highlighted a number of high-risk transactions occupying critical positions within the network.

2. Methodology Overview

The investigation followed a five-stage workflow:

- Data preparation and merging of transaction features, class labels, and edge lists
- Construction of a directed blockchain transaction graph
- Extraction of a high-risk working subgraph using centrality-driven expansion
- Computation of in-degree, out-degree, betweenness centrality, and PageRank
- Detection of layering and fan-out behavioral patterns linked to suspicious transaction movement

Metric	Value
Total Nodes	203,769
Total Edges	234,355

Metric	Value
Working Subgraph Nodes	5,459
Working Subgraph Edges	6,478

Metric	Weight
Betweenness Centrality	40%
PageRank	30%
Out-Degree	20%
In-Degree	10%

3. Three Highest-Risk Nodes

Transaction 232438397 displayed strong aggregation behavior commonly associated with collection nodes. Transaction 230457868 acted as a bridge between multiple transaction clusters, indicating possible relay or obfuscation activity. Transaction 92491280 showed unusually high systemic influence relative to its connectivity level.

Rank	Transaction ID	Classification	Risk Score	Key Observation
1	232438397	LICIT	0.544	Received 160 incoming flows into a single aggregation node
2	230457868	UNKNOWN	0.456	Highest betweenness centrality in the network
3	92491280	LICIT	0.372	High PageRank despite low direct connectivity

4 Fraud Pattern Findings

Most flagged layering nodes carried UNKNOWN classifications, indicating major monitoring gaps within the network.

Pattern	Observation
Layering Behavior	11 transactions flagged for receiving from multiple sources and redistributing funds within the same time step
Fan-Out Behavior	4 transactions flagged for dispersing value to multiple destinations rapidly
Highest Fan-Out Node	Transaction 89273 distributed funds to 288 destinations within one time step
Community Risk	Several smaller communities contained entirely unknown or suspiciously connected nodes

5. Actionable Recommendations

Recommendation	Responsible Unit
Escalate Transactions 232438397 and 230457868 for blockchain tracing and wallet attribution	FIU / Blockchain Tracing Team
Implement continuous monitoring for high-centrality unknown nodes	Compliance Monitoring Team
Conduct periodic reclassification of unknown-labelled transactions using chain analytics tools	Compliance & Intelligence Unit

6. Data Limitation

A major limitation of this analysis is the large proportion of unknown-labelled transactions within the dataset. Risk scores were derived primarily from structural network behavior rather than confirmed wallet attribution. Findings should therefore be treated as indicators of elevated compliance risk requiring further investigation, not direct evidence of criminal activity.